

ENTERPRISE RETAIL INTELLIGENCE DASHBOARD

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ENTERPRISE RETAIL INTELLIGENCE DASHBOARD

Business Intelligence & Predictive Analytics Case Study

1. PROJECT OVERVIEW

The Enterprise Retail Intelligence Dashboard is an interactive Power BI project designed to analyze retail sales, customer, product, and market performance.

The project transforms raw transactional data into meaningful business insights through data cleaning, data modeling, DAX measures, interactive dashboards, customer segmentation, RFM analysis, and predictive forecasting.

The dashboard enables business stakeholders to monitor performance, identify growth opportunities, understand customer behavior, evaluate product performance, analyze market contribution, and support future business planning.

2. BUSINESS PROBLEM

Retail businesses generate large volumes of transactional data, but raw data does not directly provide actionable business insights.

The key challenge is to convert transactional sales data into a centralized business intelligence solution that can answer important questions related to revenue, customers, orders, products, markets, customer segments, and future performance.

This project addresses the problem by developing an interactive Power BI dashboard that provides both historical performance analysis and predictive business insights.

3. BUSINESS OBJECTIVES

- Analyze overall revenue, orders, customers, and market performance.
- Identify high-value and at-risk customer segments.
- Analyze product revenue, quantity, pricing, and order performance.
- Identify the highest-performing markets.
- Understand revenue concentration across markets.
- Analyze customer behavior using RFM segmentation.
- Forecast future revenue and order performance.
- Provide actionable recommendations for customer retention, inventory readiness, and market growth.
- Create an interactive dashboard that allows stakeholders to explore performance using date filters and visual interactions.

4. DATASET & DATA PREPARATION

The project uses retail transaction-level data containing information related to invoices, products, quantities, prices, customers, countries, and invoice dates.

The dataset was prepared before analysis by cleaning and transforming the raw transactional data.

Key preparation activities included:

- Handling invalid and inconsistent records.
- Preparing date-related fields.
- Creating Month Year and Month Sort fields.
- Creating Revenue from Quantity and Price.
- Preparing customer-level data for RFM analysis.
- Creating a separate RFM customer segmentation table.
- Establishing relationships between transactional and customer segmentation data.

5. DATA MODEL

The Power BI data model connects the cleaned retail transaction table with the RFM customer segmentation table using Customer ID.

The transactional table contains multiple records for individual customers, while the RFM table contains customer-level segmentation information.

The relationship enables customer segmentation insights to be combined with transactional sales analysis while maintaining interactive filtering across the dashboard.

6. KEY DAX MEASURES

The project uses DAX measures to calculate important business KPIs and analytical metrics.

Key measures include:

- TOTAL REVENUE
- TOTAL ORDERS
- TOTAL CUSTOMERS
- TOTAL QUANTITY
- AVERAGE ORDER VALUE
- AVG REVENUE PER CUSTOMER
- AVG ORDERS PER CUSTOMER
- REVENUE PER ORDER
- TOP MARKET REVENUE
- REVENUE CONCENTRATION
- HIGH VALUE CUSTOMERS
- AT RISK CUSTOMERS
- CHAMPIONS
- POTENTIAL LOYALISTS
- PREDICTED REVENUE 6M
- PREDICTED ORDERS 6M

7. DASHBOARD STRUCTURE

PAGE 1 — EXECUTIVE SUMMARY

Provides an overall view of business performance through revenue, orders, customers, average order value, monthly revenue trends, top countries, top customers, and top products.

PAGE 2 — CUSTOMER ANALYSIS

Focuses on customer behavior and segmentation.

It includes customer segment distribution, revenue by customer segment, top customers, purchase frequency, and detailed RFM customer information.

PAGE 3 — PRODUCT PERFORMANCE

Analyzes product-level performance using revenue, quantity sold, average price, and order count.

This page helps identify high-performing products and products with strong sales volume.

PAGE 4 — MARKET PERFORMANCE

Analyzes market-level performance using revenue, orders, revenue per order, and market contribution.

This page helps identify high-performing markets and areas requiring greater business focus.

PAGE 5 — EXECUTIVE INSIGHTS & FUTURE OUTLOOK

Combines historical performance, revenue forecasting, sales and order trends, customer insights, product insights, market insights, RFM opportunities, and predictive business outlook.

The page provides management-level insights and strategic recommendations.

8. KEY BUSINESS QUESTIONS

The dashboard is designed to answer the following business questions:

- How is overall revenue performing?
- Which markets generate the highest revenue?
- Which customers contribute the most revenue?
- Which products generate the highest revenue?
- Which products have the highest sales volume?
- Which customers are high-value customers?
- Which customers are at risk?
- How concentrated is revenue across markets?
- Which customer segments have the highest purchase frequency?
- What is the expected revenue for the next six months?
- What is the expected order volume for the next six months?

9. KEY INSIGHTS

The dashboard provides insights across four major business areas:

CUSTOMER INSIGHTS

High-value customers and at-risk customers can be identified through RFM segmentation, allowing the business to prioritize retention and customer engagement strategies.

PRODUCT INSIGHTS

Revenue, quantity, average price, and order-count analysis helps identify products that drive revenue as well as products that generate high sales volume.

MARKET INSIGHTS

Market-level revenue and order analysis highlights the strongest markets and reveals the level of dependency on major revenue-generating markets.

PREDICTIVE INSIGHTS

Revenue and order forecasting provides an outlook for the next six months and supports forward-looking business planning.

10. PREDICTIVE ANALYSIS

The dashboard includes a six-month revenue forecast based on historical monthly revenue trends.

The forecast provides:

- Next 6 Months Revenue Forecast
- Expected Growth
- Average Monthly Forecast
- Confidence Interval
- Predicted Orders for the Next 6 Months
- Predicted Revenue for the Next 6 Months

The predictive analysis helps management move from historical reporting toward forward-looking decision-making.

11. BUSINESS RECOMMENDATIONS

CUSTOMER RETENTION

Focus on retaining high-value customers and reducing the risk of losing valuable customer segments through targeted engagement and loyalty strategies.

PRODUCT OPTIMIZATION

Prioritize high-revenue and high-volume products while reviewing pricing and product performance to improve profitability.

MARKET EXPANSION

Identify emerging and high-performing markets and evaluate opportunities for geographic expansion and localized marketing.

INVENTORY READINESS

Use product demand and order trends together with the predictive outlook to improve inventory planning and reduce the risk of stock shortages.

12. FINAL BUSINESS IMPACT

The dashboard provides a centralized view of retail business performance and converts complex transactional data into actionable insights.

It enables stakeholders to monitor performance, understand customer behavior, identify important products and markets, evaluate business risks, and prepare for future demand.

The combination of descriptive analytics, RFM segmentation, interactive filtering, and predictive forecasting makes the solution useful for both operational monitoring and strategic decision-making.

13. TOOLS & TECHNOLOGIES

- Power BI
- Power Query
- DAX
- Data Modeling
- RFM Analysis
- Data Visualization
- Predictive Forecasting

14. PROJECT LEARNINGS

Through this project, I developed practical experience in:

- Data cleaning and transformation
- Power BI data modeling
- Relationship management
- DAX measure development
- Customer segmentation
- RFM analysis
- KPI development
- Interactive dashboard design
- Predictive forecasting
- Business storytelling
- Converting analytical findings into business recommendations

PROJECT OUTCOME

The final solution is an interactive Enterprise Retail Intelligence Dashboard that combines historical analysis, customer intelligence, product and market performance, RFM segmentation, and predictive forecasting into a single business intelligence solution.